

Luong Phat Nguyen

PhD in Computer Vision | Senior AI R&D Engineer — Multimodal LLMs & Voice AI

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PROFILE

7+ years of AI R&D (PhD included), from real-time computer vision to **multimodal LLMs and generative voice AI**. Specialist in taking research prototypes to **production at scale**: an iOS SDK deployed in retail, a proprietary TTS engine under 200 ms, voice agents in customer service, **4 international publications**. Looking for a **Senior Machine Learning Engineer** position to drive high-business-impact multimodal and voice systems and grow an R&D team.

PROFESSIONAL EXPERIENCE

Senior AI R&D Engineer | [Rizlum AI](#)

Mar. 2025 – Present | Paris, France

- Designed multimodal document-understanding models (insurance and health mutual insurers: health-care invoices, motor-accident statements) reaching **over 95 % correct extraction**: processing cut to a **few seconds per document**.
- Built Rizlum AI's proprietary Speech-to-Text and Text-to-Speech models (**sub-200 ms latency**): conversation with no perceptible dead air, and third-party voice APIs no longer needed.
- Deployed a real-time speech-to-speech system for conversational voice agents: **95 % of membership sign-ups automated** at client MGP, with the journey available around the clock and no queueing.
- Set up synthetic data generation and identity-data anonymisation (regulated sector): training **on par with real-data performance**, with no exposure of personal data.

Technologies: Multimodal LLMs, TTS, ASR, PyTorch, Hugging Face, real-time inference

Postdoctoral Researcher — LLM | [Toyota Motor Europe & LISSI](#)

Jul. 2024 – Mar. 2025 | Paris, France

- Designed and trained multimodal LLMs (vision + language), Mamba and Transformer architectures, in an industrial partnership with the carmaker.
- Industrialised the training pipelines on large-scale multimodal datasets (VQA, Conceptual Captions).
- Created an emotionally intelligent conversational agent, leading to an **IEEE ICIP 2025 publication**.

Technologies: Multimodal LLMs, Mamba, facial emotion recognition, PyTorch

AI R&D Engineer | [Trax Retail](#)

Dec. 2021 – Apr. 2024 | Paris, France

- Developed on-shelf product recognition models (**over 95 % accuracy across more than 500 SKUs**), replacing manual in-store stocktaking with a photo audit taking a few seconds.
- Embedded real-time algorithms in Swift into an iOS SDK (CoreML, ARKit, SceneKit) shipped to the product teams: on-device inference, **with no network dependency and no server inference cost**.
- Optimised and generalised models pre-trained on large-scale image data: accuracy gains **without massive re-annotation**, reducing the cost and lead time of rolling out to new retailers and new countries.
- Computed business indicators (*share-of-shelf*) in real time at **under 1 % error and under 200 ms latency**, usable by brands during the store visit.

Technologies: PyTorch, CoreML, ARKit, Swift, OpenCV, 3D vision, augmented reality

PhD Candidate — Computer Vision | [LIFAT, University of Tours](#)

Nov. 2018 – Dec. 2021 | Tours, France

- Carried out a thesis on the extraction and characterisation of spatio-temporal patterns in videos, leading to **2 publications** at international conferences (ICPR, VISAPP).
- Designed video classification architectures (R(2+1)D, R3D, LSTM, Vision Transformers) that are **lightweight and trainable on little annotated data**.
- Evaluated and applied optical-flow estimation methods (variational / PDE-based, FlowNet2, PWC-Net) and sequential pattern mining methods (N-gram, GSP, PrefixSpan, BIDE).

Technologies: PyTorch, Keras, OpenCV, Python, C++, data mining

CORE SKILLS

AI & deep learning	Multimodal LLMs & VLMs, Speech-to-Text, Text-to-Speech, speech-to-speech, conversational agents, Transformers, CNN, LSTM, state-space models (Mamba), fine-tuning (LoRA / PEFT), model optimisation and distillation
Computer vision	Image and video recognition and classification, optical-flow estimation, augmented reality, 3D vision, spatio-temporal pattern extraction
Production	Real-time and low-latency inference, on-device models (CoreML), industrialisation of training pipelines, identity-data anonymisation and synthetic data generation (regulated sector)
Languages	Python, C/C++, Swift, Bash, JavaScript, Matlab, LaTeX
Frameworks & tools	PyTorch, Hugging Face, Keras, OpenCV, CoreML, ARKit, SceneKit, Docker, Git, Linux, Google Cloud, FastAPI
Transferable	Applied research, research → product transfer, scientific publication, university teaching, large-scale data processing

EDUCATION

PhD in Computer Vision | University of Tours – LIFAT Laboratory **2018 – 2021**

Topic: *Extraction and characterisation of spatio-temporal patterns in videos.*

Engineering Degree (MSc equivalent) | INSA Centre Val de Loire, Blois campus **2015 – 2018**

Major: *Automated systems and industrial computing.*

PUBLICATIONS

L. P. Nguyen, H. Abdelkawy, A. Othmani. *EmoMamba: Advancing Dynamic Facial Expression Recognition with Visual and Textual Fusion*. IEEE International Conference on Image Processing (**ICIP**), 2025, pp. 1432–1437.

L. P. Nguyen, J. Mille, D. H. Li, D. Conte, N. Ragot. *Efficient Dynamic Texture Classification with Probabilistic Motifs*. International Conference on Pattern Recognition (**ICPR**), 2022, pp. 564–570.

L. P. Nguyen, J. Mille, D. Li, D. Conte, N. Ragot. *Trajectory Extraction and Deep Features for Classification of Liquid-gas Flow under the Context of Forced Oscillation*. International Conference on Computer Vision Theory and Applications (**VISAPP**), 2020, pp. 17–26.

L. P. Nguyen, M. Saleh, R. Le Bouquin Jeannès. *An Efficient Design of a Machine Learning-based Elderly Fall Detector*. International Conference on IoT Technologies for HealthCare (**HealthyIoT**), Springer, 2017, pp. 34–41.

TEACHING

- **Neural Networks – Deep Learning** (lectures and lab sessions) – 5th year, Polytech Tours, University of Tours, 2020.
- **Object-Oriented Programming – C++** (lab sessions) – 3rd year, INSA Centre Val de Loire (GSI), 2019–2020.
- **Introduction to Deep Learning** (lab sessions) – MSc 2 Big Data Management and Analytics, 2018–2019.

AWARDS

- *Rencontres du Vietnam* excellence scholarship – Odon Vallet, 2016–2018.
- Silver medal, Vietnam National Physics Olympiad, 2014.

LANGUAGES

English	Fluent (advanced)
French	Fluent (professional)
Vietnamese	Native